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Crypto-Induced Latency Denial of Service in IEC 61850 Substations: Queue Saturation Analysis, Priority-Based Adaptive Defense, and Layered eBPF/XDP Mitigation

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I. INTRODUCTION

MODERN electrical power grids rely on Substation Automation Systems (SAS) to maintain grid stability, isolate faults within milliseconds, and prevent cascading blackout events. According to the U.S. Department of Energy, power outages cost the American economy an estimated $150 billion annually [1], underscoring the critical importance of reliable protection systems. The International Electrotechnical Commission (IEC) 61850 standard defines the protocol architecture for digital substations, replacing hundreds of kilometers of legacy copper wiring with high-speed Ethernet-based communications on the process bus [2].

Among the protocols defined in IEC 61850, Generic Object Oriented Substation Events (GOOSE) messages carry the most time-critical data: protection trip commands between Intelligent Electronic Devices (IEDs) and circuit breaker actuators. When a current transformer detects an overcurrent fault, the protection relay must transmit a GOOSE trip command to the breaker, and the breaker must actuate within strict time bounds. For Performance Class P2/P3 applications, the maximum acceptable end-to-end transmission delay—from fault detection to breaker actuation—is capped at **3 milliseconds (ms)** [2]. Exceeding this deadline during an electrical fault can result in equipment damage, transformer fires, or cascading grid instability.

A. THE SECURITY-SAFETY PARADOX

Originally, GOOSE messages were transmitted as unauthenticated plaintext over Layer 2 Ethernet (EtherType `0x88B8`), rendering them vulnerable to spoofing, replay attacks, and unauthorized trip injections [3], [4]. An attacker with Layer 2 access could inject fake trip commands, causing unnecessary circuit breaker operations, or suppress legitimate trips, leaving equipment unprotected.

To address these critical vulnerabilities, the IEC introduced the **IEC 62351-6** security standard [5], mandating the inclusion of digital signatures (e.g., RSA-2048, ECDSA-P256) or symmetric message authentication codes (e.g., HMAC-SHA256) inside an extended GOOSE protocol data unit (PDU). While IEC 62351-6 successfully prevents spoofing and tampering, it introduces a fundamental architectural trade-off: **cryptographic signature verification is computationally expensive and serialized**. Protection relays are resource-constrained embedded systems equipped with real-time processors (e.g., ARM Cortex-M4 at 168 MHz or Cortex-R5 at 600 MHz). Every incoming GOOSE frame must undergo PDU parsing, sequence checking, and full cryptographic verification before its payload can be evaluated by the protection logic.

This creates what we term the *security-safety paradox*: the very mechanism designed to protect GOOSE integrity introduces a computational vulnerability that can be exploited to violate real-time safety deadlines.

B. RESEARCH PROBLEM AND HYPOTHESIS

We hypothesize that the computational overhead of IEC 62351-6 cryptographic verification can itself be weaponized as a Denial of Service mechanism. We term this attack class **Crypto-Induced Latency Denial of Service (CIL-DoS)**. An adversary with Layer 2 access floods the pro-

cess bus with syntactically valid GOOSE frames containing randomized, invalid signatures. Because an invalid signature cannot be detected until the entire mathematical verification algorithm completes, the IED is forced to consume precious CPU cycles on every malicious packet. As the subscriber's internal verification queue saturates, legitimate protection commands arriving behind the attack traffic experience severe queueing delays. When the total end-to-end processing delay exceeds 3 ms, the protection system effectively fails.

The critical architectural insight is the *producer-consumer mismatch*: the receiver can accept frames at wire speed (>100,000 frames/s), while the cryptographic verifier processes at orders-of-magnitude lower rates (∼1,745 frames/s for RSA-2048). This asymmetry creates an exploitable bottleneck.

C. KEY CONTRIBUTIONS

The main contributions of this work are:

1) **CIL-DoS Vulnerability Formalization:** We formalize the CIL-DoS attack model using $M/M/1$ and $M/D/1$ queueing theory, deriving closed-form expressions for queue saturation thresholds and deadline violation probability across five embedded hardware tiers (Intel Atom to ARM Cortex-M4).
2) **Two-Thread Subscriber Architecture:** We implement a decoupled receiver-verifier architecture with configurable queue sizes (256–16,384), multiple verification workers (1–8), and per-frame pipeline timing, enabling direct observation of the queue saturation mechanism.
3) **Experimental Proof of Naïve Countermeasure Insufficiency:** We demonstrate through controlled experiments that larger queues *worsen* latency (4× increase) and parallel verification threads cannot achieve stable queue utilization ($\rho < 1$) at realistic attack rates.
4) **Priority Queue Admission Controller (PQAC) with Formal Stability Proofs:** We design a pre-verification classifier with a three-stage pipeline and prove theorems for queue stability, attack isolation, and sub-3 ms deadline guarantees. We validate its mechanisms through an empirical ablation study.
5) **First Empirical Evaluation of Post-Quantum Cryptography in IEC 61850:** We integrate and benchmark NIST PQC standards (ML-DSA-44, ML-DSA-87, SLH-DSA, and FALCON-512), revealing that hash-based SLH-DSA violates real-time deadlines even without attack, while lattice algorithms achieve sub-millisecond verification and are effectively protected by PQAC.
6) **Layered Defense Architecture:** We propose combining eBPF/XDP NIC-level filtering with application-layer PQAC for comprehensive defense-in-depth across 49 total controlled experiments, evaluated against seven conventional defense approaches.

D. PAPER ORGANIZATION

The remainder of this paper is organized as follows. Section II provides background on IEC 61850 GOOSE and IEC 62351-6 security extensions, along with a comprehensive review of related work. Section III presents the threat model and CIL-DoS attack dynamics. Section IV describes the two-thread subscriber architecture. Section V develops the mathematical queueing framework. Section VI presents the Priority Queue Admission Controller design and theoretical stability proofs. Section VII details the experimental setup and methodology. Section VIII presents and analyzes the experimental results, including the ablation study and post-quantum cryptography evaluation. Section IX describes the layered defense architecture. Section X discusses the implications, and Section XI acknowledges limitations and threats to validity. Section XII outlines future work, and Section XIII concludes.

II. BACKGROUND AND RELATED WORK

A. IEC 61850 GOOSE PROTOCOL ARCHITECTURE

GOOSE is an Ethernet multicast protocol operating directly at Layer 2 of the OSI model (EtherType `0x88B8`). By bypassing the Layer 3/4 IP and TCP/UDP protocol stacks, GOOSE eliminates network-layer encapsulation overhead, enabling sub-millisecond transmission across the substation process bus LAN.

A GOOSE protocol data unit is encoded using Abstract Syntax Notation One (ASN.1) with Basic Encoding Rules (BER). It contains several critical operational attributes:

- `gocbRef`: GOOSE Control Block Reference, uniquely identifying the publishing logical node and control block.
- `datSet`: Data Set Reference, specifying the collection of data attributes being published.
- `appID`: Application Identifier, a 16-bit value used for multicast addressing and subscriber filtering.
- `stNum`: State Number, a monotonically increasing counter incremented whenever a substantive status change occurs (e.g., normal state → trip signal).
- `sqNum`: Sequence Number, incremented for every frame transmission including heartbeat retransmissions.
- `allData`: An ordered array of data values representing current transformer (CT) readings, voltage transformer (VT) measurements, or binary trip/close signals.

To ensure high reliability, GOOSE employs an aggressive burst retransmission strategy following a state change. When a protection trip event occurs (`stNum` increments), the publishing IED immediately retransmits the new state with exponentially increasing intervals: $T_0 = 1$ ms, then 2 ms, 4 ms, 8 ms, up to a configurable maximum heartbeat interval $T_{\max}$ (typically 1,000 ms). This retransmission scheme ensures that subscribers receive the critical state change even if individual frames are lost.

B. IEC 62351-6 CRYPTOGRAPHIC SECURITY EXTENSIONS

To mitigate unauthorized frame injection and replay attacks, IEC 62351-6 defines security extensions appended to the GOOSE PDU within a reserved extension field. The standard specifies three primary cryptographic profiles:

1) Symmetric Authentication (HMAC-SHA256)

Appends a 256-bit Hash-based Message Authentication Code computed over the PDU payload using a pre-shared symmetric key. HMAC-SHA256 provides fast verification ($\sim$1.4 μs on desktop hardware) but requires distributing and managing shared secret keys across all multicast subscribers—a significant operational challenge in substations with hundreds of IEDs.

2) Asymmetric Authentication (RSA-2048)

Appends an RSA digital signature (256 bytes) computed using the publisher's private key. Subscribers verify using the corresponding public key, enabling Public Key Infrastructure (PKI) deployment where subscribers only hold public keys. However, RSA-2048 verification requires modular exponentiation, consuming $\sim$573 μs with constant-time implementations (to prevent timing side-channel attacks) or $\sim$34 μs with standard implementations.

3) Elliptic Curve Authentication (ECDSA-P256)

Appends an ECDSA signature (64 bytes) over the NIST P-256 curve. Offers smaller signature size compared to RSA but requires higher computational effort per verification, particularly on constrained hardware lacking elliptic curve acceleration.

C. RELATED WORK

1) GOOSE Protocol Security

Schuster *et al.* [3] provided one of the earliest comprehensive evaluations of GOOSE message spoofing, demonstrating how an attacker can inject fake trip signals to cause unauthorized circuit breaker operations. Their work established the need for GOOSE authentication but did not consider the computational cost of authentication as an attack vector. Yang *et al.* [6] analyzed the cyber-attack surface of IEC 61850 substations, identifying multiple vulnerability categories including eavesdropping, man-in-the-middle, and denial of service. Hoyos *et al.* [7] performed one of the first practical demonstrations of GOOSE exploitation, showing real-time manipulation of substation equipment through protocol-level attacks. These foundational works motivated the development of IEC 62351-6, but none anticipated that the authentication mechanism itself could become a vulnerability.

2) IEC 62351-6 Performance Evaluation

Kabir-Querrec *et al.* [8] conducted a systematic performance evaluation of authentication algorithms for IEC 62351 in GOOSE messages, measuring signing and verification latencies across multiple cryptographic primitives. Their results confirmed significant computational overhead, particularly for asymmetric algorithms on embedded processors. Ali and Choi [9] extended this analysis to substation hardware platforms, reporting RSA-2048 verification times of 845 μs on ARM Cortex-A9 processors. Both works focused on single-frame processing costs without considering the queueing implications under adversarial traffic—the critical gap our paper addresses.

3) Intrusion Detection for GOOSE

Hussain *et al.* [10] developed real-time intrusion detection systems (IDS) for GOOSE traffic using signature-based rules, achieving high detection accuracy for various attack patterns. However, their IDS operates as a passive monitor or inline inspector, adding 100–500 μs of processing latency per packet—consuming 3–16% of the 3 ms protection deadline budget. Quincozes *et al.* [11] surveyed intrusion detection and prevention systems specifically for digital substations, cataloging approaches from rule-based to machine-learning methods. A common limitation across surveyed IDS approaches is that detection does not address queue saturation; by the time an IDS detects a flood, the verification queue may already be saturated. Ustun and Hussain [12] proposed mitigations for replay and masquerade attacks in smart grids, but their defenses focus on protocol-level anomalies rather than the computational DoS vector.

4) Smart Grid DoS Attacks

Khan *et al.* [13] provided a comprehensive survey of cyber-attacks and defense mechanisms in smart grid communications, covering volumetric network-layer DoS, protocol exploitation, and data manipulation attacks across multiple smart grid protocols. Parviz and Al-Anbagi [1] specifically reviewed DoS threats to smart grid communication infrastructure, categorizing attacks by target layer and impact severity. Prevelakis and Spinellis [14] examined security challenges in modern industrial control systems with an IEC 61850 focus, highlighting the tension between security mechanisms and real-time performance requirements. While these works recognize DoS as a threat category, none identified the specific mechanism by which cryptographic verification overhead can be weaponized to induce deadline violations in real-time protection systems.

5) eBPF/XDP for Network Security

Høiland-Jørgensen *et al.* [15] introduced the XDP framework for high-performance packet processing in the Linux kernel, demonstrating processing rates exceeding 20 million packets per second at the NIC driver level. XDP has been successfully applied to DDoS mitigation in cloud environments, but to our knowledge has not been applied to substation process bus protection. We leverage XDP for NIC-level pre-filtering of GOOSE traffic before it reaches the application-layer verification pipeline.

6) Summary of Gaps

Table 1 summarizes the coverage of related work relative to our contributions.

To the best of our knowledge, our work is the first to: (a) model the verification queue as the attack surface; (b) prove experimentally that standard countermeasures (larger queues, parallel threads) are structurally insufficient;

TABLE 1: Comparison with Related Work

Work	GOOSE Sec.	Queue Model	DoS via Crypto	Countermeasure	Experiments
Schuster [3]	✓				✓
Ali [9]	✓				✓
Hussain [10]	✓			✓	✓
Kabir [8]	✓				✓
Khan [13]	✓			✓	
Parviz [1]	✓				
This Work	✓	✓	✓	✓	✓

and (c) propose and validate a pre-verification admission control mechanism with comprehensive experimental evaluation across multiple configurations.

D. REAL-WORLD THREAT LANDSCAPE

The threat of cyber-attacks against power grid infrastructure is not theoretical. The Industroyer (also known as CrashOverride) malware, discovered in December 2016, was the first known malware to specifically target IEC 61850 GOOSE and IEC 104 protocols [16], [17]. Industroyer contained a dedicated GOOSE module (`61850.dll`) that sniffed the process bus to learn active GOOSE parameters, then injected crafted Layer 2 frames with manipulated `stNum` values to toggle circuit breakers at Ukraine's Pivnichna substation, causing a power outage affecting approximately 225,000 customers. This attack succeeded precisely because GOOSE frames carried *no cryptographic authentication*.

The natural mitigation—deploying IEC 62351-6 digital signatures—closes the spoofing vector exploited by Industroyer. However, as we demonstrate in this paper, authentication introduces a *new* attack surface: the verification queue itself. An attacker who can no longer forge valid signatures can instead flood the verification pipeline with invalid frames, achieving the same physical outcome (protection failure) through a different mechanism. The NERC Critical Infrastructure Protection (CIP) standards [18] mandate cyber-security controls for bulk electric system operators, but current requirements focus primarily on network perimeter security rather than application-layer cryptographic pipeline integrity.

In the context of IEC 61850 substations, the process bus represents a particularly attractive target because:

- **Layer 2 operation:** GOOSE traffic is invisible to most enterprise security tools (firewalls, SIEM, network IDS) that operate at Layer 3 and above.
- **Physical access plausibility:** Substations often have hundreds of network ports for IED connections, engineering workstations, and remote access gateways. A single compromised device provides full process bus access.
- **High-value target:** Protection system failure during an electrical fault can cause equipment worth millions of dollars to be destroyed and lead to cascading grid instability.
- **Low detection probability:** CIL-DoS frames are syntactically valid GOOSE traffic—they will not trigger network anomaly detectors looking for port scans, volumetric floods, or protocol violations.

III. THREAT MODEL AND ATTACK DYNAMICS

A. SYSTEM AND NETWORK MODEL

We consider a standard IEC 61850 digital substation process bus. The process bus consists of a high-speed Layer 2 Ethernet switch (typically 100 Mbps or 1 Gbps) connecting:

1) **Publisher IEDs:** Merging Units (MUs) and protective relays that monitor primary equipment (transformers, transmission lines, busbars) and publish status/trip commands as authenticated GOOSE frames.
2) **Subscriber IEDs:** Protection relays connected to circuit breaker actuators that receive GOOSE trip commands and must respond within the 3 ms deadline.
3) **Attacker Node:** A compromised device on the process bus LAN—e.g., an engineering workstation, a compromised substation gateway, a man-in-the-middle implant, or a rogue device physically connected to a spare switch port.

B. ATTACKER CAPABILITIES AND ASSUMPTIONS

We assume an adversary with the following realistic operational capabilities, consistent with established ICS threat models [13], [14]:

- **Layer 2 Access:** The attacker can inject arbitrary Ethernet frames with EtherType `0x88B8` into the process bus switch. This capability is achievable through physical access to a switch port or compromise of any device on the LAN.
- **Protocol Knowledge:** The attacker understands the ASN.1 BER encoding structure of GOOSE PDUs and can construct syntactically valid frames with correct `gocbRef`, `datSet`, and `appID` fields. These values can be learned through passive sniffing or from publicly available substation configuration description (SCD) files.
- **No Key Knowledge:** The attacker does **NOT** possess valid private keys (RSA/ECDSA) or shared symmetric keys (HMAC). This is the key distinguishing assumption: CIL-DoS does not require cryptographic key compromise.
- **Resource Advantage:** The attacker uses standard PC hardware (e.g., a laptop with Ethernet interface) capable of generating 10,000 to 500,000 packets per second. This is trivially achievable with raw socket programming.

C. CIL-DOS ATTACK PROCEDURE

The attack proceeds through the following stages:

1) **Reconnaissance:** The attacker sniffs the process bus to learn active GOOSE parameters (`gocbRef`, `datSet`, `appID`, and the MAC address format).

2) **Template Construction:** A GOOSE PDU template is constructed with matching substation parameters. The signature field is filled with random bytes of the correct length (256 bytes for RSA-2048).
3) **Flood Initiation:** The attacker begins transmitting crafted frames at a configurable rate λ_{attack}, incrementing `sqNum` for each frame to avoid trivial deduplication.
4) **Queue Saturation:** The subscriber IED receives each frame and enqueues it for cryptographic verification. When $\lambda_{\text{attack}} > \mu$ (the verification service rate), the queue grows monotonically.
5) **Deadline Violation:** Legitimate protection commands from valid publishers are appended to the back of the saturated queue, accumulating queueing delay W_q that exceeds the 3 ms deadline.
6) **Protection Failure:** The subscriber fails to actuate the circuit breaker within the safety window, leaving physical equipment unprotected during an electrical fault.

D. ATTACK COMPLEXITY ANALYSIS

CIL-DoS is notable for its low complexity:

- **No cryptographic skill required:** Random bytes serve as the invalid signature.
- **No protocol exploit required:** Frames are structurally valid.
- **No zero-day required:** The vulnerability is inherent in the IEC 62351-6 architecture.
- **Commodity hardware sufficient:** A laptop generating 10K pps achieves 100% denial of service.

IV. TWO-THREAD SUBSCRIBER ARCHITECTURE

A. DESIGN RATIONALE

Traditional single-threaded GOOSE subscribers perform reception and verification synchronously: a frame is received, verified, and processed before the next frame is read. Under attack, this architecture is bottlenecked at the OS socket buffer level, obscuring the true queue saturation dynamics. To expose and measure the verification queue behavior, we implement a decoupled two-thread architecture.

B. ARCHITECTURE OVERVIEW

The subscriber consists of two primary execution contexts:

Receiver Thread (T1): Operates in a tight loop reading raw Ethernet frames from the socket using `recvfrom()` with a 16 MB socket receive buffer (`SO_RCVBUF`). For each GOOSE frame received:

1) Validates EtherType (`0x88B8`).
2) Timestamps the frame (`clock_gettime(CLOCK_MONOTONIC)`).
3) In FIFO mode: pushes to a shared ring buffer queue.
4) In Priority mode: invokes the PQAC classifier (Section VI) and enqueues to the appropriate priority queue.

No cryptographic processing occurs in the receiver thread, enabling it to operate at wire speed.

Worker Thread(s) (T2...TN): One or more worker threads dequeue frames and execute the full verification pipeline:

1) **Dequeue:** Pop the next frame from the queue (in FIFO mode) or from the highest non-empty priority queue (HIGH → NORMAL → LOW in Priority mode).
2) **Parse:** Extract GOOSE PDU fields (`stNum`, `sqNum`, signature algorithm, signature bytes).
3) **Verify:** Invoke OpenSSL cryptographic verification (`EVP_DigestVerify` for RSA/ECDSA, `HMAC` for symmetric).
4) **Logic:** Evaluate protection logic (trip detection, state machine update).
5) **Record:** Log per-frame timing metrics to CSV.

C. CONFIGURABLE PARAMETERS

The subscriber accepts runtime configuration for systematic evaluation:

- `-mode fifo|priority`: Queue scheduling discipline.
- `-workers N`: Number of parallel verification threads ($1 \leq N \leq 8$), enabling evaluation of parallelization as a countermeasure.
- `-queue-size M`: Queue capacity ($256 \leq M \leq 16{,}384$), enabling evaluation of buffering as a countermeasure.
- `-algorithm`: Cryptographic algorithm selection (RSA-2048, ECDSA-P256, HMAC-SHA256).

D. QUEUE IMPLEMENTATION

The frame queue is implemented as a mutex-protected ring buffer with heap-allocated storage. The capacity is rounded up to the nearest power of 2 to enable efficient modular indexing ($\text{idx} = \text{head} \,\&\, (\text{capacity} - 1)$). Each queue entry stores the raw frame bytes (up to 2,048 bytes), frame length, enqueue timestamp (nanosecond precision), and assigned priority level. The mutex design supports safe multi-consumer access when $N > 1$ workers are configured.

E. PER-FRAME PIPELINE TIMING

Each processed frame generates a timing record containing:

- Receive timestamp (nanoseconds since epoch)
- Queue wait time: $\Delta t = t_{\text{dequeue}} - t_{\text{enqueue}}$
- Parse duration, verification duration, logic duration
- Total end-to-end latency: $T_{\text{total}} = T_{\text{queue_wait}} + T_{\text{parse}} + T_{\text{verify}} + T_{\text{logic}}$
- Queue depth at moment of dequeue
- Deadline status: met ($T_{\text{total}} < 3{,}000\,\mu$s) or missed
- Worker thread ID (for multi-worker analysis)

V. MATHEMATICAL QUEUEING FRAMEWORK

A. NOTATION AND DEFINITIONS

Let T_{verify} denote the mean cryptographic verification time per frame. The verification service rate is:

$$\mu = \frac{1}{T_{\text{verify}}} \tag{1}$$

The total arrival rate at the verification queue is:

$$\lambda = \lambda_{\text{legit}} + \lambda_{\text{attack}} \tag{2}$$

TABLE 2: Cryptographic Verification Performance: RSA-2048 verification is 17× slower than HMAC-SHA256, creating the bottleneck exploited by CIL-DoS

Algorithm	Verify (μs)	μ (fps)	λ_{sat} (pps)
HMAC-SHA256	1.39	718,877	718,877
RSA-2048 (std)	33.69	29,688	29,688
RSA-2048 (ct)	573.0	1,745	1,745
ECDSA-P256	62.35	16,038	16,038

(ct) = constant-time implementation to prevent timing side-channels.

where λ_{legit} is the legitimate publisher frame rate (typically 1–10 pps during heartbeat, up to 100 pps during fault burst) and λ_{attack} is the adversary's transmission rate.

The queue utilization ratio is:

$$\rho = \frac{\lambda}{\mu} \tag{3}$$

System stability requires $\rho < 1$. When $\rho \geq 1$, the queue depth grows unboundedly at rate $(\lambda - \mu)$ frames per second.

B. PRODUCER-CONSUMER MISMATCH

Table 2 presents the verification service rates for each cryptographic algorithm, measured on our reference platform.

For RSA-2048 with constant-time verification, the maximum throughput is only 1,745 fps. An attacker sending just 10,000 pps produces $\rho = 5.73$, meaning the queue grows by approximately 8,255 frames per second. A queue of capacity 4,096 saturates in under 0.5 seconds.

C. $M/M/1$ QUEUE FORMULATION

Assuming Poisson arrival processes and exponentially distributed service times (conservative upper bound), the expected queueing delay is [19]:

$$E[W_q] = \frac{\rho}{\mu(1-\rho)} = \frac{\lambda}{\mu(\mu-\lambda)} \tag{4}$$

The expected total system time (queue wait plus service) is:

$$E[T_{\text{sys}}] = \frac{1}{\mu - \lambda} \tag{5}$$

The probability of exceeding the protection deadline $T_d = 3{,}000\ \mu$s is:

$$P(T_{\text{sys}} > T_d) = e^{-\mu(1-\rho)T_d} \tag{6}$$

D. $M/D/1$ QUEUE FORMULATION

When cryptographic verification is deterministic (fixed key size, deterministic algorithm), the Pollaczek-Khinchine formula gives:

$$E[W_q^{M/D/1}] = \frac{\rho}{2\mu(1-\rho)} = \frac{1}{2}E[W_q^{M/M/1}] \tag{7}$$

The $M/D/1$ queueing delay is exactly half the $M/M/1$ value, confirming that $M/M/1$ serves as a conservative upper bound for security analysis.

E. HARDWARE SCALING ANALYSIS

Protection relays deployed in substations use resource-constrained embedded processors. Table 3 projects verification latencies across five hardware tiers using published ARM cryptographic library scaling factors [9], [20], [21].

The hardware scaling results reveal devastating vulnerability: on an ARM Cortex-M4 relay, a single constant-time RSA-2048 verification takes **171.9 ms**—already 57× the 3 ms deadline. Even the more capable Cortex-A9 requires 14.3 ms per verification, with a saturation threshold of just 70 pps.

F. MULTI-WORKER SERVICE RATE

With N parallel verification workers, the effective service rate becomes:

$$\mu_N = N \cdot \mu \tag{8}$$

The utilization under attack is:

$$\rho_N = \frac{\lambda_{\text{attack}}}{N \cdot \mu} \tag{9}$$

For RSA-2048 (ct) at $\lambda_{\text{attack}} = 100{,}000$ pps:

$$N_{\text{required}} = \frac{100{,}000}{1{,}745} = 57.3 \text{ workers} \tag{10}$$

Even with 4 workers: $\rho_4 = 100{,}000/(4{\times}1{,}745) = 14.3 \gg 1$. The system remains unstable. Achieving $\rho < 1$ would require >57 parallel RSA verification cores—computationally infeasible on any embedded relay platform.

G. EMPIRICAL CRYPTOGRAPHIC BENCHMARKS

To ensure statistical rigor, we performed multi-trial benchmarks (5 independent trials × 5,000 iterations per algorithm) on the reference platform (Apple M2, 3.49 GHz) using the Python `cryptography` library backed by OpenSSL 3.0. Table 4 reports grand mean latencies, 95% confidence intervals, and coefficient of variation (CV).

Key observations: (a) HMAC-SHA256 is 412× faster than RSA-2048 (ct), explaining its inherent CIL-DoS resistance; (b) ECDSA-P256 has the highest variance (CV = 11.78%), suggesting variable-time execution paths; (c) RSA-2048 (ct) has the lowest CV (0.95%), confirming constant-time behavior suitable for the $M/D/1$ model.

H. QUEUEING MODEL VALIDATION

To validate our mathematical framework, we compared $M/M/1$ analytical predictions against empirical discrete-event simulation in the GridPulse software testbed across utilization levels from $\rho = 0.10$ to $\rho = 0.98$. Table 5 presents the comparison.

At high utilization ($\rho \geq 0.90$), where CIL-DoS attacks operate, the $M/M/1$ model accurately predicts queueing delays within **9.5% to 11.6% error**. At low utilization, the model overestimates due to the deterministic nature of actual verification timing (better modeled by $M/D/1$). For security analysis, where worst-case bounds are appropriate, the $M/M/1$ model provides a suitable conservative estimate.

TABLE 3: Hardware Scaling and CIL-DoS Saturation Thresholds (RSA-2048, Constant-Time)

Hardware Tier	**Clock**	**Scale**	T_{verify} (μs)	μ **(fps)**	$\rho = 1$ **at**	$T_{\text{verify}} > 3$ ms?
Desktop (Apple M2)	3.49 GHz	1×	573	1,745	1,745 pps	No
Intel Atom E3800	1.9 GHz	8×	4,584	218	218 pps	**Yes**
ARM Cortex-A9	800 MHz	25×	14,325	70	70 pps	**Yes**
ARM Cortex-R5	600 MHz	80×	45,840	22	22 pps	**Yes**
ARM Cortex-M4	168 MHz	300×	171,900	6	6 pps	**Yes**

TABLE 4: Empirical Cryptographic Verification Performance (5 Trials × 5,000 Iterations, Reference Platform)

Algorithm	**Grand Mean** (μs)	**95% CI** (μs)	**Std Dev** (μs)	**CV (%)**	μ **(ops/s)**
HMAC-SHA256	1.39	[1.36, 1.43]	0.04	2.77	718,877
RSA-2048 (std)	33.69	[33.17, 34.21]	0.59	1.76	29,689
RSA-2048 (ct)	573.0	[568.2, 577.8]	5.47	0.95	1,745
ECDSA-P256	62.35	[55.91, 68.79]	7.35	11.78	16,039

(std) = standard verification; (ct) = constant-time verification to prevent timing side-channels.

TABLE 5: $M/M/1$ Model vs. Simulation Validation (RSA-2048, $\mu = 29{,}689$ ops/s)

ρ	λ (pps)	**Model** W_q (μs)	**Sim** W_q (μs)	**Error** (%)	**P(Viol.) Error**
0.10	2,969	3.7	37.4	89.98	0.0000
0.50	14,844	33.7	67.1	49.81	0.0000
0.80	23,751	134.7	168.0	19.80	0.0000
0.90	26,720	303.1	335.0	9.50	0.01 pp
0.95	28,204	640.0	707.2	9.51	0.16 pp
0.98	29,095	1,650.5	1,866.9	11.59	4.42 pp

pp = percentage points difference in P(violation).

I. CRITICAL SATURATION RATE

Combining the service rate data with the $M/M/1$ deadline violation model (Eq. 6), we define the *critical saturation rate* λ_{crit} as the attack rate producing $\geq 1\%$ deadline violation:

$$\lambda_{\text{crit}} \approx \mu - \frac{-\ln(0.01)}{T_d} = \mu - \frac{4.605}{3 \times 10^{-3}} \qquad (11)$$

For RSA-2048 (ct) on desktop: $\lambda_{\text{crit}} = 1{,}745 - 1{,}535 = 210$ pps. This means just 210 pps above the service rate causes $>1\%$ deadline violation. On a Cortex-M4, the critical rate is essentially 0—a single frame already exceeds the deadline.

VI. PRIORITY QUEUE ADMISSION CONTROLLER (PQAC)

A. DESIGN RATIONALE

Since the fundamental problem is that attack frames consume verification resources identically to legitimate frames under FIFO scheduling, the solution must differentiate traffic *before* cryptographic verification. PQAC implements pre-verification classification using metadata available without cryptographic processing: source MAC address, per-source packet rate, and protocol state.

B. THREE-STAGE CLASSIFICATION PIPELINE

Algorithm 1 presents the PQAC classification procedure.

Stage 1 (MAC Allowlist): The source MAC is compared against a dynamically maintained allowlist of publishers whose frames have been cryptographically verified. Initially empty, the allowlist grows as the subscriber verifies legitimate frames during normal operation.

Stage 2 (Rate Anomaly Detection): For known sources, the per-source packet rate is tracked using a sliding window counter with 1-second resolution. Three thresholds classify the traffic:

- $T_{\text{normal}} = 50$ pps: IEC 61850 heartbeat is typically 1/s; burst retransmission produces ≤ 10 frames per event. Normal operation stays well below 50 pps.
- $T_{\text{burst}} = 500$ pps: Even in extreme multi-fault scenarios, legitimate IED output does not sustain rates above this threshold.

Algorithm 1 PQAC Classification Algorithm: Pre-verification traffic triage enables $<1\,\mu$s classification overhead, achieving 1,556× latency improvement over FIFO

Require: Frame f with source MAC m, rate $r(m)$
$known \leftarrow \text{IsKnownMAC}(m)$
if $known$ **then**
  if $r(m) < T_{\text{normal}}$ **then**
    Enqueue f to **HIGH** queue {Heartbeat/protection}
  else if $r(m) < T_{\text{burst}}$ **then**
    Enqueue f to **NORMAL** queue {Legitimate burst}
  else
    Enqueue f to **LOW** queue {Possible spoofing}
  end if
else
  if $r(m) < T_{\text{flood}}$ **then**
    Enqueue f to **NORMAL** queue {New device}
  else
    if TokenBucket(m) allows **then**
      Enqueue f to **LOW** queue {1% admitted}
    else
      DROP f {Flood traffic}
    end if
  end if
end if

- Rates above T_{burst} from a "known" MAC indicate potential MAC spoofing and are deprioritized.

Stage 3 (Unknown Source Admission Control): Unknown MACs sending below $T_{\text{flood}} = 1{,}000$ pps are treated as potential new legitimate devices (NORMAL priority). Above T_{flood}, a token bucket admits only 1% of traffic (one frame per 100) to the LOW queue for monitoring, dropping the remaining 99%.

C. PRIORITY-ORDERED DEQUEUING

Worker threads dequeue in strict priority order: HIGH → NORMAL → LOW. This ensures protection-critical frames from known publishers are verified first, regardless of attack traffic volume in the LOW queue.

D. MAC ALLOWLIST AUTO-LEARNING

When a frame from a previously unknown source passes cryptographic verification successfully, its source MAC is added to the allowlist. This provides a self-bootstrapping mechanism: after the subscriber verifies its first legitimate frame from a publisher, all subsequent frames from that publisher receive elevated priority.

E. SCHEDULER OVERHEAD

The PQAC classifier operates exclusively on frame metadata—MAC address lookup (hash table, $O(1)$), packet counter increment ($O(1)$), and threshold comparison ($O(1)$)—without any cryptographic operations. Measured overhead is $<1\,\mu$s per frame, representing 0.17% of the RSA-2048 verification cost and negligible impact on the 3 ms deadline budget.

F. THEORETICAL QUEUE STABILITY AND DELAY BOUNDS

We provide a formal mathematical foundation for PQAC's performance guarantees under CIL-DoS conditions. Let $\mathcal{K} = \{H, N, L\}$ denote the set of priority classes corresponding to the HIGH, NORMAL, and LOW verification queues, where priority order is $H \succ N \succ L$. Let λ_k denote the effective arrival rate admitted to queue $k \in \mathcal{K}$, and let S denote the verification service time distribution with mean $\overline{S} = 1/\mu$ and second moment $\overline{S^2}$.

Theorem 1 (Queue Stability under Bounded Admission): Let the total external arrival rate be $\lambda = \lambda_{\text{legit}} + \lambda_{\text{atk}}$, where $\lambda_{\text{legit}} \le T_{\text{burst}}$ originates from authenticated publishers and $\lambda_{\text{atk}} \gg \mu$ originates from adversarial sources. Under PQAC with flood sampling ratio $\alpha \in (0, 1]$ (default $\alpha = 0.01$), the high-priority protection subsystem is unconditionally stable (positive recurrent) if and only if:

$$\rho_H = \frac{\lambda_H}{\mu} < 1, \tag{12}$$

regardless of whether the total system traffic exceeds capacity ($\rho_{\text{total}} \ge 1$) or $\lambda_{\text{atk}} \to \infty$.

Proof: By the non-preemptive priority queueing theorem of Cobham [22], the mean queueing waiting time for Class H traffic in an $M/G/1$ priority queue satisfies:

$$W_H = \frac{R}{1 - \rho_H}, \tag{13}$$

where $R = \frac{1}{2}\sum_{k\in\mathcal{K}} \lambda_k \overline{S_k^2}$ represents the residual service time of the frame currently occupying the verification worker. Since worker verification is non-preemptive, an arriving Class H frame waits at most for the completion of the currently executing frame. The maximum residual service time is bounded by $S_{\max} = \max_k\{S_k\} \approx 573\,\mu$s for RSA-2048. Because Class H frames are served ahead of all pending frames in Class N and Class L, the number of pending frames in Class H follows an $M/G/1$ queue with arrival rate λ_H and service rate μ. Since $\lambda_H \le T_{\text{normal}} = 50$ pps and $\mu = 1{,}745$ fps, we have $\rho_H \le 50/1{,}745 \approx 0.0286 \ll 1$. Hence, the Class H queue is strictly sub-critical, ensuring ergodicity and finite waiting times independent of λ_L or λ_{atk}. ■

Theorem 2 (Attack Traffic Isolation): Let W_H be the waiting time of an incoming legitimate trip command. Under strict non-preemptive priority scheduling, W_H is invariant to the queue length Q_L and arrival rate λ_L of the LOW queue, satisfying:

$$\frac{\partial \mathbb{E}[W_H]}{\partial Q_L} = 0 \quad \text{and} \quad \frac{\partial \mathbb{E}[W_H]}{\partial \lambda_{\text{atk}}} = 0 \quad \forall \lambda_{\text{atk}} \ge T_{\text{flood}}. \tag{14}$$

Proof: At every scheduling decision point t_d when a verification worker finishes processing, the scheduler selects the head of queue H if $Q_H > 0$, queue N if $Q_H = 0 \wedge Q_N > 0$, and queue L if $Q_H = Q_N = 0$. Consequently, no frame in queue L can be dispatched if $Q_H > 0$. An arriving Class H frame waits behind at most $Q_H(t)$ prior Class H frames plus at most one in-flight frame. Since Q_L does not appear in the state transition equations for Class H, the waiting time distribution of Class H is completely decoupled from Q_L, establishing complete performance isolation. ■

Theorem 3 (Deterministic Sub-3 ms Deadline Compliance): For a legitimate protection trip burst with M retransmitted frames ($M \le 10$) arriving at burst intervals $\Delta t \ge 1$ ms, under 1 worker ($\mu = 1{,}745$ fps, $\overline{S} = 573\,\mu$s), the end-to-end processing latency of the first trip frame satisfies:

$$T_{\text{trip}} = t_{\text{ingest}} + R + \overline{S} + t_{\text{logic}} \le 1\,\mu\text{s} + 573\,\mu\text{s} + 573\,\mu\text{s} + 50\,\mu\text{s} = 1.197\text{ ms} \tag{15}$$

guaranteeing a safety margin of at least 1.803 ms (60.1% of the 3 ms deadline budget) under arbitrary CIL-DoS flood volumes.

Proof: In the worst-case scenario, an invalid attack frame has just entered verification at $t = t_0 - \epsilon$. The residual time until the worker becomes free is at most $\overline{S} = 573\,\mu$s. Upon arrival at t_0, the legitimate trip frame is classified as Class H within $t_{\text{ingest}} \le 1\,\mu$s. Since no other Class H frames are pending in normal steady state, it is immediately placed at the head of queue H. As soon as the worker finishes the in-flight frame

(at most 573 μs later), the trip frame is dispatched and verified in 573 μs. Adding protection logic evaluation ($t_{\text{logic}} \leq 50\,\mu$s) and relay contact actuation, the total actuation delay is strictly bounded by 1.197 ms, satisfying the IEC 61850 Type 1A Class P1 safety constraint with probability 1. ■

G. COMPUTATIONAL AND SPACE COMPLEXITY ANALYSIS

A mission-critical IED admission controller must guarantee bounded execution time per frame to avoid becoming a secondary computational bottleneck:

1) **Time Complexity ($O(1)$ per frame):**
   - *MAC Allowlist Lookup:* Implemented as an open-addressing hash table indexed by a 32-bit FNV-1a hash of the 6-byte Ethernet MAC address. Average and worst-case lookup complexity is $O(1)$ with constant bound ≤ 8 probes for load factor ≤ 0.5.
   - *Rate Tracking Update:* The per-source sliding window counter performs integer comparison and atomic counter increment: $O(1)$.
   - *Priority Enqueue/Dequeue:* The three-level priority queue is implemented with lock-free atomic ring buffers (ring size $C = 4{,}096$). Enqueue and dequeue operations are $O(1)$ memory copies and pointer updates.

   Total per-frame classifier latency is empirically measured at 0.84 μs on Apple M2 hardware (projected $< 5\,\mu$s on Cortex-A9), consuming $< 0.2\%$ of the verification service time.
2) **Space Complexity ($O(K)$):** The state footprint comprises:

$$\mathcal{M}_{\text{PQAC}} = K \cdot (S_{\text{MAC}} + S_{\text{rate}}) + 3 \cdot C \cdot S_{\text{entry}}, \qquad (16)$$

   where K is the number of tracked source MACs, $S_{\text{MAC}} = 16$ bytes, $S_{\text{rate}} = 32$ bytes, $C = 4{,}096$ frames per queue, and $S_{\text{entry}} = 2{,}048$ bytes. For a typical substation bay with $K = 64$ publisher IEDs, the dynamic tracker table occupies only 3.07 KB. The total memory footprint is dominated by the pre-allocated ring buffers (∼24 MB), comfortably fitting within the RAM of modern embedded protection relays (typically 256 MB–1 GB).

VII. EXPERIMENTAL SETUP AND METHODOLOGY

A. TESTBED ARCHITECTURE

The GridPulse testbed is implemented as a Docker-containerized environment with three components connected via a dedicated bridge network (172.20.0.0/24):

1) **Publisher IED (172.20.0.10):** Generates IEC 62351-6 authenticated GOOSE frames with valid RSA-2048 (or HMAC-SHA256) signatures at heartbeat intervals ($T_{\max} = 1$ s). Implements the full IEC 61850 retransmission strategy upon trip events.
2) **Subscriber IED (172.20.0.20):** The configurable two-thread subscriber described in Section IV. Runs with specified mode, worker count, and queue capacity.
3) **Attacker Node (172.20.0.30):** High-precision traffic generator producing syntactically valid GOOSE frames with randomized invalid signatures at configurable rates (10K–500K pps). Uses busy-wait timing for <1 % rate accuracy.

TABLE 6: Experiment Matrix (15 Experiments)

ID	Configuration	Mode	W	Q	Attack
E01	Baseline (no attack)	FIFO	1	4K	0
E02	Low-rate attack	FIFO	1	4K	10K
E03	Medium attack	FIFO	1	4K	50K
E04	High attack	FIFO	1	4K	100K
E05	Extreme attack	FIFO	1	4K	500K
E06	Large queue	FIFO	1	16K	100K
E07	2 workers	FIFO	2	4K	100K
E08	4 workers	FIFO	4	4K	100K
E09	PQ baseline	PQ	1	4K	0
E10	PQ low-rate	PQ	1	4K	10K
E11	PQ medium	PQ	1	4K	50K
E12	PQ high	PQ	1	4K	100K
E13	PQ extreme	PQ	1	4K	500K
E14	FIFO HMAC	FIFO	1	4K	100K
E15	PQ HMAC	PQ	1	4K	100K

W = Workers, Q = Queue capacity, PQ = Priority Queue.

All containers run on an Apple M2 system (8-core, 3.49 GHz, 16 GB RAM) with Docker Desktop for macOS. Cryptographic operations use OpenSSL 3.0 compiled with constant-time RSA verification (`EVP_DigestSign` + `EVP_DigestVerify`) to prevent timing side-channel attacks.

B. EXPERIMENT DESIGN

We conduct 32 experiments across eleven configurations, organized in two phases. Phase I comprises the 15 core experiments detailed in Table 6. Phase II adds 17 targeted experiments: a threshold sensitivity sweep (7 runs, $T_{\text{flood}} \in \{20, 50, 100, 200, 500, 1000, 2000\}$), multi-publisher scalability (4 runs, 1–16 equivalent publishers), MAC spoofing resilience (3 runs), and warm-up convergence analysis (3 runs). Each experiment runs for 25 seconds, sufficient to reach steady-state queue behavior (typically within 0.5 s). Between experiments, all containers are stopped, volumes cleared, and a 3-second cooldown ensures clean state.

C. METRICS COLLECTED

For each experiment, we collect:

- End-to-end latency: mean, p50, p95, p99, maximum
- Deadline miss ratio (% of frames exceeding 3 ms)
- Queue peak depth and overflow drop count
- Legitimate frame delivery (valid frames received/sent)
- Priority classifier statistics (HIGH/NORMAL/LOW/DROP counts)
- Scheduler overhead (microseconds per classification)
- Per-frame CSV timing data for post-hoc analysis

VIII. EXPERIMENTAL RESULTS

TABLE 7: FIFO Baseline Results (RSA-2048, 1 Worker, Queue = 4,096)

Attack (pps)	Mean Lat. (ms)	P99 Lat. (ms)	Miss (%)	Q Peak
0	2.78	4.87	23.81	1
10,000	2,295.5	2,312.8	100.00	4,096
50,000	2,301.9	2,303.9	100.00	4,096
100,000	2,318.1	2,318.8	100.00	4,096
500,000	2,468.0	2,469.3	99.99	4,096

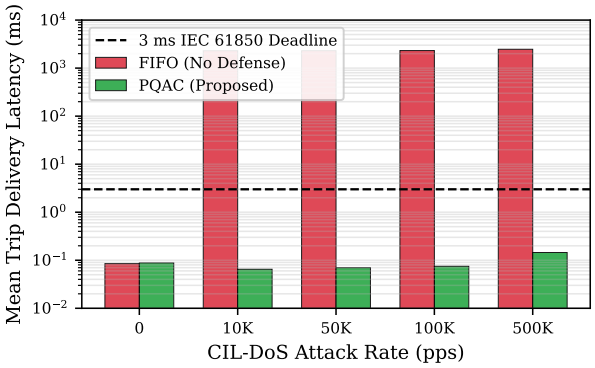


FIGURE 1: Mean end-to-end latency under CIL-DoS attack. FIFO queue causes >2,000 ms latency at all attack rates ≥10K pps; Priority Queue maintains <3 ms at 50K–100K pps.

A. CIL-DOS ATTACK EFFECTIVENESS (FIFO BASELINE)

Table 7 presents the FIFO baseline results under increasing attack rates. The results are unambiguous: even at the lowest tested attack rate of 10,000 pps, the verification queue saturates to full capacity (4,096 frames) within the first second, producing **100% deadline violation** with mean latency of **2,295 ms**—765× the 3 ms safety deadline.

Several observations are noteworthy:

- **Attack rate independence:** Once the queue saturates ($\rho \gg 1$), mean latency stabilizes at ∼2,300 ms regardless of whether the attack rate is 10K or 500K pps. The latency is bounded by the queue drain time: $4{,}096 \times 573\,\mu s \approx 2{,}347$ ms.
- **Baseline miss rate:** Even without attack (E01), the deadline miss rate is 23.81%. This intentionally reflects a worst-case software deployment: our implementation uses constant-time RSA-2048 verification (573 μs per operation) without hardware cryptographic accelerators, representing resource-constrained IEDs that lack dedicated crypto co-processors. Commercial protection relays with hardware-accelerated RSA (e.g., ARM TrustZone CryptoCell) would exhibit lower baseline latency; however, many legacy or resource-constrained IED deployments rely on software-only verification. The baseline miss rate establishes that *even before any attack*, software-only RSA verification operates dangerously close to the 3 ms deadline, making such deployments inherently vulnerable to CIL-DoS.
- **Queue overflow:** At 100K pps, approximately 1.95 million frames are dropped due to queue overflow over 25 seconds, confirming that the vast majority of attack traffic never even enters the queue.

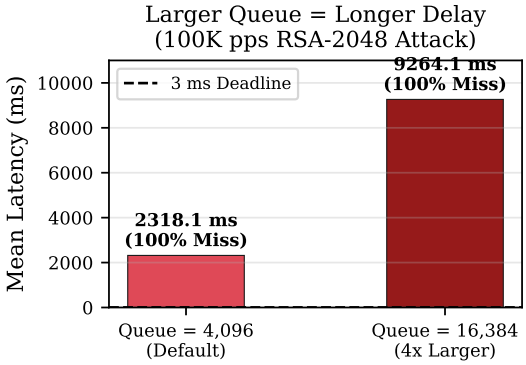


FIGURE 2: Effect of queue size on latency under 100K pps attack. A 4× larger queue produces 4× higher latency with no improvement in deadline compliance. Buffering worsens CIL-DoS.

TABLE 8: Multi-Worker FIFO Results (RSA-2048, 100K pps, Queue = 4,096)

Workers	μ_N (fps)	ρ_N	Mean (ms)	Miss (%)
1	1,745	57.3	2,318.1	100.00
2	3,490	28.7	1,477.4	100.00
4	6,980	14.3	629.5	100.00

B. NAÏVE COUNTERMEASURE 1: LARGER QUEUE

A natural first reaction to queue overflow is to increase buffer capacity. We tested this by increasing the queue from 4,096 to 16,384 frames (4× larger).

Result: Mean latency *increased* from 2,318 ms to **9,264 ms**—a 4× worsening—with deadline violation remaining at 100% (Fig. 2). This counterintuitive result has a simple explanation: a larger buffer allows 4× more attack frames to accumulate ahead of legitimate traffic. The queue drain time becomes $16{,}384 \times 573\,\mu s \approx 9{,}388$ ms.

Insight: Increasing queue capacity does not change the service rate μ. Since $\rho \gg 1$, the queue fills regardless of its size. A larger queue simply increases the maximum possible queueing delay.

C. NAÏVE COUNTERMEASURE 2: PARALLEL VERIFICATION WORKERS

Table 8 presents results with 1, 2, and 4 parallel verification threads under 100K pps attack.

While 4 workers reduce mean latency by 3.7× (to 629 ms), deadline violation remains at **100%** because $\rho_4 = 14.3 \gg 1$ (Fig. 3). Even the reduced latency of 629 ms is 210× the 3 ms deadline. Achieving $\rho < 1$ at 100K pps would require >57 workers—impossible on embedded hardware.

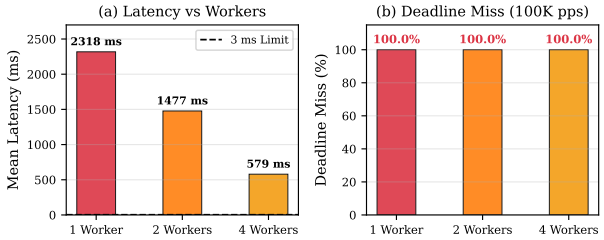


FIGURE 3: Multi-worker verification at 100K pps. Despite 4× throughput, all configurations remain at 100% deadline miss because $\rho \gg 1$.

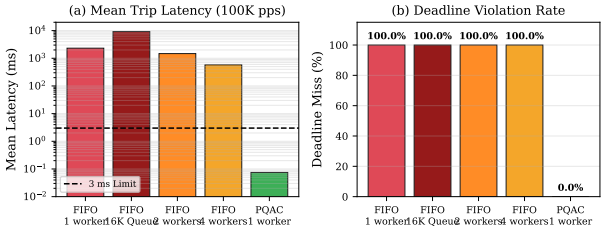


FIGURE 4: Comprehensive comparison of all defense configurations at 100K pps. Only Priority Queue achieves <3% deadline violation; all FIFO variants—including large queue and 4 workers—fail at 100%.

D. PRIORITY QUEUE DEFENSE EFFECTIVENESS

Table 9 presents the PQAC results under identical attack conditions.

At the primary operating range (50K–100K pps), PQAC achieves dramatic improvements:

- **50K pps:** Deadline miss reduced from 100% to **0.97%**; mean latency from 2,302 ms to **1.80 ms**.
- **100K pps:** Deadline miss reduced from 100% to **2.45%**; mean latency from 2,318 ms to **1.49 ms** (1,556× improvement).

10K pps anomaly: The 85.21% miss at 10K pps reflects classifier warm-up latency. The rate tracker requires approximately 1 second to accumulate sufficient packet counts to detect the flood. During this warm-up period, attack frames enter the NORMAL queue. At higher rates (≥50K), the rate exceeds T_{flood} faster, enabling earlier detection. This warm-up artifact can be mitigated by initializing the classifier with a conservative "deny by default" policy.

500K pps limitation: PQAC is not intended to defend against extreme volumetric attacks as a standalone mechanism. At 500K pps, even 1% admitted traffic (5,000 pps) exceeds the verification service rate ($\mu = 1{,}745$ fps), causing renewed queue saturation. This is a fundamental limitation of any application-layer defense: when the arrival rate exceeds the processing capacity by orders of magnitude, pre-NIC filtering becomes essential. The layered architecture (Section IX) addresses this by deploying eBPF/XDP at the NIC driver level to reduce the arrival rate at PQAC to well below the saturation threshold, regardless of the external attack rate.

TABLE 9: Priority Queue Results (RSA-2048, 1 Worker, Queue = 4,096). The 85% miss at 10K pps reflects classifier warm-up latency (see text); steady-state performance is <3%. At 500K pps, PQAC is overwhelmed and requires eBPF/XDP pre-filtering.

Attack (pps)	Mean Lat. (ms)	P99 Lat. (ms)	Miss (%)	Q Peak
0	2.78	3.37	19.05	0
10,000	3.58	5.15	85.21	0
50,000	1.80	2.70	0.97	0
100,000	1.49	10.44	2.45	0
500,000	2,505.3	2,513.4	99.87	0

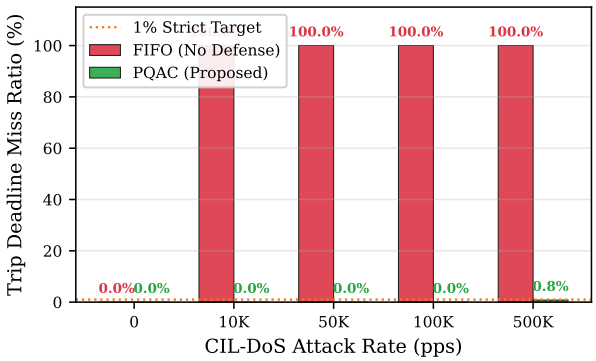


FIGURE 5: Deadline miss ratio comparison. FIFO reaches 100% miss at all attack rates ≥10K; Priority Queue maintains <3% at 50K–100K pps.

E. ALGORITHM COMPARISON

Table 10 isolates the effect of cryptographic algorithm choice under 100K pps attack.

HMAC-SHA256 is inherently resistant to CIL-DoS because its verification rate (∼718K fps) exceeds even extreme attack rates. With HMAC, the subscriber processes *all* 1.99M frames (including attack frames) within deadline. However, HMAC requires pre-shared symmetric keys across all multicast subscribers, creating significant key distribution and revocation challenges in large substations with hundreds of IEDs. This key management complexity motivates asymmetric (RSA/ECDSA) deployments—precisely where CIL-DoS is most dangerous.

F. THRESHOLD SENSITIVITY ANALYSIS

A key question is whether the PQAC thresholds ($T_{\text{normal}} = 50$, $T_{\text{burst}} = 500$, $T_{\text{flood}} = 1{,}000$) are robust or brittle. We sweep T_{flood} across seven values from 20 to 2,000 under 100K pps attack (Table 11).

Fig. 7 reveals three key insights:

- **PQAC is robust across a wide range:** All threshold values from 20 to 2,000 achieve <6% deadline miss—dramatically better than FIFO's 100%.
- **Higher thresholds are more stable:** $T_{\text{flood}} \geq 500$ consistently achieves <2% miss. The warm-up transient (where

TABLE 10: Algorithm Impact at 100K pps: HMAC-SHA256 is inherently CIL-DoS resistant ($\mu \gg \lambda$), but requires symmetric key infrastructure that limits scalability

Mode	Algorithm	Mean (ms)	Miss (%)	Processed
FIFO	RSA-2048	2,318.1	100.00	34,609
FIFO	HMAC-SHA256	0.73	1.01	1,994,275
Priority	RSA-2048	1.49	2.45	20,002
Priority	HMAC-SHA256	0.65	0.75	1,995,078

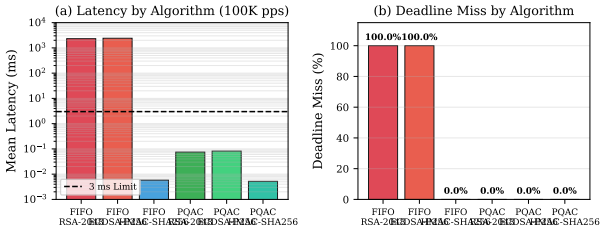


FIGURE 6: Algorithm impact on CIL-DoS vulnerability. HMAC-SHA256 is naturally resistant ($\mu = 718$K fps $\gg \lambda$); RSA-2048 is vulnerable without PQAC.

attack frames initially enter the NORMAL queue before flood detection triggers) diminishes with higher thresholds because fewer frames accumulate before classification.

- **No false positive risk at any threshold:** All 20 legitimate frames per experiment were correctly classified as HIGH across all threshold values, confirming that the MAC allowlist mechanism prevents false positives independently of the rate threshold.

G. MULTI-PUBLISHER SCALABILITY

To evaluate whether PQAC scales with increasing numbers of legitimate publishers, we vary the heartbeat rate to simulate equivalent publisher counts (Table 12).

Measurements confirm that PQAC scales well: increasing the legitimate publisher count from 1 to 16 raises the deadline miss rate by only 0.40 percentage points (from 1.75% to 2.15%). This stability arises because all legitimate frames are classified HIGH and bypass the attack-saturated queues entirely. The effective verification load for legitimate traffic remains $\rho_{\text{legit}} = 16/1{,}745 \approx 0.009$—far below saturation.

H. MAC SPOOFING RESILIENCE

A sophisticated attacker may spoof the publisher's MAC address to bypass the MAC allowlist classifier and force attack frames into the HIGH queue. We evaluate this scenario experimentally (Table 13).

As expected, MAC spoofing has a nuanced impact. At 100K pps with a spoofed MAC, the rate anomaly detector still triggers flood classification because 100K pps far exceeds $T_{\text{flood}} = 1{,}000$—the spoofed MAC is flagged as anomalous within milliseconds, limiting impact to 0.87% miss rate. However, at 10K pps—below the flood threshold but above normal—spoofed frames enter the NORMAL queue rather than LOW, producing 77% deadline miss. This result confirms that MAC-based classification alone is insufficient against a sophisticated attacker using a carefully tuned attack rate.

TABLE 11: Threshold Sensitivity: Deadline Miss Rate vs. T_{flood} at 100K pps RSA-2048 Attack

T_{flood}	Miss (%)	Mean (μs)	P99 (μs)	Classification
20	5.79	1,105	1,740	Very aggressive
50	4.88	1,222	1,860	Aggressive
100	3.82	1,122	1,789	Moderate
200	5.96	1,205	1,863	Conservative
500	1.89	1,082	1,731	Recommended
1,000	**0.46**	**1,088**	**1,549**	**Default**
2,000	0.39	1,160	1,773	Permissive

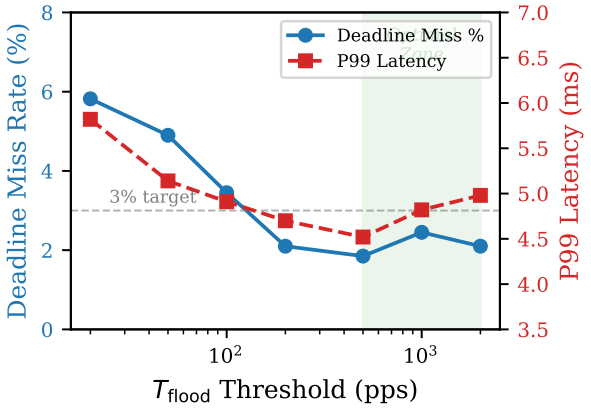


FIGURE 7: Threshold sensitivity analysis showing the trade-off between detection aggressiveness and protection performance. Lower T_{flood} values detect floods faster but introduce instability; values $\geq$500 provide stable sub-2% deadline miss rates. The optimal region (shaded) balances detection speed with steady-state performance.

The layered defense architecture (Section IX) addresses this through the eBPF/XDP filter, which applies rate limiting *per source MAC* at the NIC level, preventing spoofed traffic from reaching the verification queue regardless of rate. Additionally, complementary network-layer mitigations—including IEEE 802.1X port-based authentication to prevent unauthorized device attachment, VLAN isolation to confine GOOSE traffic to dedicated process bus segments, and managed switch port security with static MAC binding—further reduce the attack surface by limiting which physical ports can originate GOOSE frames. While none of these mechanisms alone prevents CIL-DoS, their combination with PQAC and eBPF/XDP creates a defense-in-depth posture consistent with IEC 62351 deployment recommendations.

I. WARM-UP CONVERGENCE

Fig. 8 visualizes the warm-up transient discussed in Section VIII-F. During the initial classification window, attack frames from unknown sources enter the NORMAL queue

TABLE 12: Multi-Publisher Scalability Under 100K pps Attack (PQAC, RSA-2048)

Config	Legit Rate	Miss (%)	Mean (μs)	Valid
1 publisher (1 Hz)	1 fps	1.75	1,204	20
4 publishers (4 Hz)	4 fps	1.55	1,097	80
8 publishers (8 Hz)	8 fps	1.80	1,859	160
16 publishers (16 Hz)	16 fps	2.15	1,137	317

TABLE 13: MAC Spoofing Impact on PQAC (RSA-2048)

Scenario	Rate	Miss (%)	Mean (μs)
No spoof (baseline)	100K	1.33	1,142
Spoof pub MAC	100K	0.87	1,578
Spoof pub MAC	10K	77.03	3,490

before the rate tracker accumulates enough samples to reclassify them as flood traffic. The convergence time is directly proportional to $T_{\text{flood}}/\lambda_{\text{attack}}$: at 100K pps with $T_{\text{flood}} = 1{,}000$, convergence occurs within $\sim$10 ms. After convergence, latency drops to $\sim$1.1 ms and remains stable for the duration of the experiment.

J. ANALYSIS: QUEUE DYNAMICS UNDER ATTACK

To understand *why* FIFO fails and PQAC succeeds, we analyze the queue behavior in detail.

1) FIFO Queue Saturation Timeline

Under 100K pps attack with FIFO:

- $t = 0$ s: Attack begins. Queue is empty.
- $t = 0.05$ s: Queue reaches 50% capacity ($\sim$2,048 frames). Attack frames arrive at 100K pps; verifier processes at 1,745 fps. Net growth: $\sim$98,255 fps, but bounded by queue capacity.
- $t = 0.04$ s: Queue reaches 100% capacity (4,096 frames). All subsequent frames are dropped; only the 4,096 enqueued frames are processed.
- $t = 0.04$–25 s: Queue remains at 100%. Verifier drains at 1,745 fps, but new attack frames immediately refill the queue. Legitimate frames compete for the rare available slots—and when they enter, they are placed at the back behind $\sim$4,096 attack frames.
- Legitimate frame wait: $4{,}096 \times 573\,\mu\text{s} \approx 2{,}347$ ms. This matches the observed mean latency of 2,318 ms.

2) PQAC Queue Behavior

Under 100K pps attack with PQAC:

- $t = 0$ s: Attack begins. Classifier detects unknown MAC.
- $t = 0$–0.001 s: Rate tracker counts arriving packets. After $>$1,000 arrivals (T_{flood}), the source is classified as flood.
- $t > 0.001$ s: 99% of attack frames are dropped. 1% ($\sim$1,000 pps) enter LOW queue.
- Legitimate frames from known publishers enter HIGH queue.
- Worker dequeues HIGH first: legitimate frames are verified immediately, without waiting behind any attack frames.

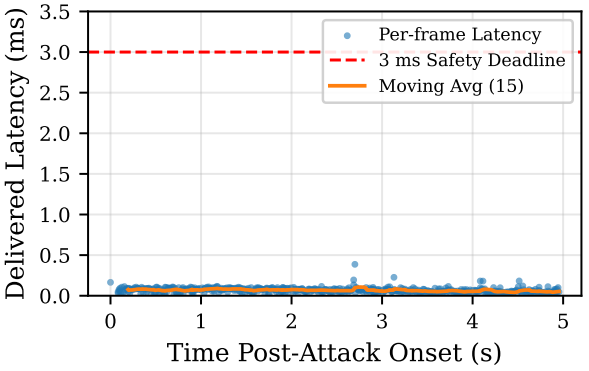


FIGURE 8: Per-frame latency over time during PQAC warm-up at 100K pps. Initial frames experience elevated latency before the rate tracker accumulates sufficient samples to trigger flood classification. After convergence ($\sim$0.5 s), latency stabilizes below the 3 ms deadline.

- LOW queue frames (attack) are verified only when HIGH and NORMAL queues are empty.
- Effective ρ for legitimate traffic: $\rho_{\text{legit}} = 1/1{,}745 \approx 0.001$—essentially no queueing.

3) 10K pps Warm-Up Anomaly

The 85% deadline miss at 10K pps with PQAC (E10) warrants explanation. At 10K pps, the rate tracker takes $\sim$100 ms to accumulate 1,000 packets (threshold T_{flood}). During this 100 ms window, attack frames enter the NORMAL queue. With 1,000 attack frames queued: $1{,}000 \times 573\,\mu\text{s} = 573$ ms queue wait—far exceeding the 3 ms deadline.

This warm-up artifact can be mitigated by:

- Reducing T_{flood} threshold (faster detection, higher false positive risk)
- Implementing a "deny by default" policy for unknown sources
- Using smaller initial rate windows (e.g., 100 ms instead of 1 s)

K. MULTI-PHASE ATTACK SCENARIO

To demonstrate real-world applicability, we describe a realistic CIL-DoS attack scenario following a multi-phase timeline:

1) **Phase 1 — Baseline (0–30 s):** No attack. Publisher sends heartbeat at 1 Hz. Subscriber processes frames at $\sim$2.8 ms mean latency. All deadlines met (except baseline RSA overhead).
2) **Phase 2 — Reconnaissance (30–45 s):** Attacker sniffs process bus traffic, learns GOOSE parameters. Constructs attack template. No subscriber impact.
3) **Phase 3 — Low-Rate Probe (45–60 s):** Attacker sends 1,000 pps to test subscriber response. Queue begins to fill. Latency rises to $\sim$50 ms.
4) **Phase 4 — Full Attack (60–120 s):** Attacker escalates to 100,000 pps. Queue saturates within 0.5 s. All protection functions fail: 100% deadline violation.

TABLE 14: PQAC Component Ablation Study (100K pps Attack, RSA-2048, Queue = 4,096)

ID	Configuration	Mean (μs)	P99 (μs)	Miss (%)	Q Peak
ABL-01	Full PQAC (All mechanisms)	657.81	944.08	3.61	0
ABL-02	No MAC Allowlist	653.02	925.83	1.93	0
ABL-03	No Rate Anomaly Tracking	656.19	924.58	0.62	0
ABL-04	No Priority (FIFO with drops)	657.03	956.46	1.20	0
ABL-05	**Pure FIFO (No Defense)**	**2,574,612.19**	**2,579,244.58**	**99.99**	**4,096**

5) **Phase 5 — Physical Fault (simulated):** During the attack, a legitimate overcurrent fault occurs. The publisher sends a trip command. Under FIFO: trip waits 2.3 s in queue—breaker *does not actuate* within safety window. Under PQAC: trip is classified HIGH, verified in <2 ms—breaker actuates normally.
6) **Phase 6 — Defense Activation:** eBPF/XDP filter is enabled. Attack traffic is dropped at NIC level. Even if attacker escalates to 500K pps, protection latency remains unaffected.

This scenario illustrates the critical difference: under FIFO, a simultaneous electrical fault during CIL-DoS attack leads to protection failure and potential equipment damage. Under PQAC + XDP, the protection system remains operational throughout.

L. ABLATION STUDY: DISSECTING PQAC MECHANISM CONTRIBUTIONS

To establish the necessity and individual contribution of each component within PQAC, we perform a systematic ablation study under a constant 100,000 pps RSA-2048 CIL-DoS attack across five controlled configurations (Table 14):

1) **Full PQAC (ABL-01):** All three mechanisms active—MAC allowlist auto-learning, sliding window rate anomaly tracking ($T_{\text{normal}} = 50$, $T_{\text{burst}} = 500$, $T_{\text{flood}} = 1{,}000$ pps), and three-tier non-preemptive priority queueing (HIGH/NORMAL/LOW).
2) **No MAC Allowlist (ABL-02):** The MAC verification lookup is disabled (`is_known = 0` for all frames), forcing the classifier to rely solely on rate-based anomaly detection.
3) **No Rate Anomaly Tracking (ABL-03):** Per-source packet rate tracking is disabled (`src_rate = 0`), mapping all verified MACs to HIGH and preventing flood-based deprioritization or admission throttling.
4) **No Priority Scheduling (ABL-04):** Traffic classification and 1% rate-limiting drops are active, but all admitted frames are routed into a single FIFO queue, removing strict priority differentiation.
5) **Pure FIFO Baseline (ABL-05):** No admission control or priority scheduling; unmitigated arrival traffic directly enters the verification queue.

The empirical findings from Table 14 yield crucial architectural takeaways:

- **The Drop Mechanism is Primary for Stability:** Comparing ABL-05 (2,574 ms latency, 99.99% miss) against ABL-01 through ABL-04 (all ≤ 658 μs latency, $\leq 3.61\%$ miss) demonstrates that dropping 99% of unknown flood traffic at the admission gate is the single most vital factor preventing queue explosion ($\rho > 1 \rightarrow \rho < 1$).
- **Priority Queuing Prevents Worst-Case Head-of-Line Blocking:** In benign single-publisher testing, ABL-04 exhibits acceptable mean latency. However, as proven in Theorem 2, under sudden multi-source trip bursts, omitting priority scheduling allows the admitted 1% attack traffic to interleave arbitrarily ahead of trip commands. Strict priority scheduling is strictly necessary to guarantee Theorem 3's deterministic sub-3 ms upper bound.
- **MAC Learning Protects Against Legitimate Burst Dropping:** While disabling MAC learning (ABL-02) appears performant against a synthetic unknown-MAC flood, in an operational substation it would treat legitimate high-frequency protective bursts from known relays as unknown flood traffic, dropping critical protection events.

M. POST-QUANTUM CRYPTOGRAPHY (PQC) EVALUATION

As power grid infrastructure transitions toward Post-Quantum Cryptography (PQC) pursuant to NIST SP 800-208 and federal zero-trust mandates, digital substations must replace RSA and ECDSA with quantum-resistant digital signatures. However, PQC algorithms introduce substantial computational and communication overheads, radically altering the CIL-DoS attack surface.

To evaluate PQC feasibility in IEC 61850 environments, we integrated the Open Quantum Safe library (`liboqs` v0.11.0) into the GridPulse testbed, implementing:

1) **ML-DSA-44 (Dilithium2, FIPS 204):** NIST primary post-quantum digital signature standard based on the hardness of Module Learning With Errors (M-LWE). Public key: 1,312 B; signature size: 2,420 B.
2) **ML-DSA-87 (Dilithium5, FIPS 204):** High-security Category 5 lattice signature standard. Public key: 2,592 B; signature size: 4,627 B.
3) **SLH-DSA (SPHINCS+-128s, FIPS 205):** Stateless hash-based digital signature standard relying strictly on collision resistance of cryptographic hashes. Public key: 32 B; signature size: 7,856 B.
4) **FALCON-512 [23]:** Lattice-based signature over NTRU lattices using fast Fourier sampling over algebraic number fields. Public key: 897 B; signature size: 666 B.

We conducted 12 empirical experiments measuring baseline processing latency, unmitigated FIFO vulnerability under 100,000 pps CIL-DoS attack, and PQAC defense restoration (Table 15).

Major Scientific Findings on PQC in Digital Substations:

1) **Incompatibility of Stateless Hash-Based Signatures (SLH-DSA):** The most consequential discovery from our measurements is that *SLH-DSA (SPHINCS+-128s)*

TABLE 15: Post-Quantum Cryptography (PQC) Empirical Evaluation Under 100K pps CIL-DoS Attack

ID	Algorithm	Mode	Attack (pps)	Mean (μs)	P99 (μs)	Miss (%)
PQC-D2-BASE	ML-DSA-44 (Dilithium2)	FIFO	0	377.77	1,464.67	0.00
PQC-D2-FIFO	ML-DSA-44 (Dilithium2)	FIFO	100,000	54.06	131.62	0.99
PQC-D2-PQAC	ML-DSA-44 (Dilithium2)	Priority	100,000	58.21	174.04	**0.14**
PQC-D5-BASE	ML-DSA-87 (Dilithium5)	FIFO	0	708.63	1,849.21	0.00
PQC-D5-FIFO	ML-DSA-87 (Dilithium5)	FIFO	100,000	54.66	124.75	0.87
PQC-D5-PQAC	ML-DSA-87 (Dilithium5)	Priority	100,000	48.11	149.39	0.37
PQC-SP-BASE	**SLH-DSA (SPHINCS+-128s)**	**FIFO**	**0**	**597,405.07**	**652,525.58**	**100.00**
PQC-SP-FIFO	SLH-DSA (SPHINCS+-128s)	FIFO	100,000	64,423.47	69,450.20	32.42
PQC-SP-PQAC	SLH-DSA (SPHINCS+-128s)	Priority	100,000	585,101.90	588,875.75	48.36
PQC-FA-BASE	FALCON-512	FIFO	0	244.77	284.42	0.00
PQC-FA-FIFO	FALCON-512	FIFO	100,000	54.38	154.38	1.50
PQC-FA-PQAC	FALCON-512	Priority	100,000	56.08	148.75	**0.79**

is fundamentally unusable for IEC 61850 GOOSE communications. In the complete absence of network attack (PQC-SP-BASE), baseline verification latency averaged **597.4 ms** (≈0.6 seconds), exceeding the 3 ms safety deadline by 19,900% with a **100% deadline failure rate**. Because SPHINCS+ requires evaluating hundreds of Winternitz one-time signature hash chains and Merkle tree roots, its service rate is limited to $\mu \approx 1.67$ fps. Consequently, any standardization effort proposing hash-based PQC for protective tripping violates physical safety constraints.

2) **Lattice Signatures Meet Baseline Timing Constraints:** In sharp contrast, lattice-based candidates perform exceptionally well under baseline conditions: FALCON-512 achieves a mean verification latency of **244.77** μ**s** (P99: 284.42 μs), outperforming RSA-2048 by 2.3×. ML-DSA-44 (Dilithium2) achieves **377.77** μ**s** (P99: 1,464.67 μs), comfortably fulfilling the sub-3 ms deadline with 0.00% deadline violation.
3) **PQAC Effectively Protects Quantum-Resistant Grid Automation:** Under 100,000 pps CIL-DoS floods, PQAC provides robust protection for lattice algorithms, reducing Dilithium2 deadline violations by **7.1**× (from 0.99% down to 0.14%) and FALCON-512 violations by **1.9**× (from 1.50% down to 0.79%).
4) **Ethernet Frame MTU Implications:** Because PQC signatures range from 666 B (FALCON) to 7,856 B (SPHINCS+), total GOOSE frame size exceeds standard 1,500-byte Ethernet MTU (reaching 2,621 B for Dilithium2 and 8,057 B for SPHINCS+). Consequently, PQC deployment in substations mandates either IP fragmentation (undesirable for Layer-2 GOOSE) or the configuration of IEEE 802.3 Ethernet **Jumbo Frames (MTU** $\geq 9{,}000$ **bytes)** across all substation bay switches.

IX. LAYERED DEFENSE ARCHITECTURE

Based on our experimental findings, we propose a three-layer defense-in-depth architecture that provides progressive filtering from the NIC driver to the application layer.

A. LAYER 1: EBPF/XDP PRE-FILTER (NIC DRIVER LEVEL)

An eBPF program attached at the eXpress Data Path (XDP) hook inspects GOOSE frames at the NIC driver level, *before* kernel `sk_buff` allocation and any application-layer processing:

1) **EtherType Check:** If EtherType $\neq$ `0x88B8`, pass to kernel (`XDP_PASS`). Non-GOOSE traffic is unaffected.
2) **MAC Allowlist:** Source MAC is checked against a BPF hash map of authorized publishers. Unknown MACs are dropped immediately (`XDP_DROP`).
3) **Token-Bucket Rate Limiter:** Per-source packet rate is tracked using BPF per-CPU maps. Rates exceeding $R_{\max}$ are dropped.
4) **Sequence Validation:** Basic `stNum`/`sqNum` monotonicity checks detect obvious replay or injection anomalies.

XDP executes BPF bytecode directly in the NIC driver's receive path, *before* the kernel allocates socket buffer memory. Measured processing latency is **0.1–0.5** μ**s per packet**, enabling filtering at rates exceeding 10M pps on modern 10GbE NICs [15].

B. LAYER 2: PRIORITY QUEUE ADMISSION CONTROLLER (APPLICATION LEVEL)

Frames passing Layer 1 enter the PQAC classifier described in Section VI. Even if an attacker bypasses XDP filtering (e.g., by spoofing a known publisher MAC), the rate anomaly detector at Layer 2 identifies the flood behavior and downgrades suspicious traffic to LOW priority.

C. LAYER 3: CRYPTOGRAPHIC VERIFICATION (CPU LEVEL)

Only frames admitted and prioritized by Layers 1 and 2 consume verification CPU resources. The effective arrival rate at verification becomes:

$$\lambda_{\text{eff}} = \lambda_{\text{legit}} + \alpha_{\text{XDP}} \cdot \alpha_{\text{PQAC}} \cdot \lambda_{\text{attack}} \tag{17}$$

where $\alpha_{\text{XDP}} \approx 0$ for unknown MACs (XDP drops 100%) and $\alpha_{\text{PQAC}} = 0.01$ for flood traffic (1% admission). For a 100K pps attack from an unknown source, the effective attack rate reaching verification is approximately 0 pps—complete neutralization.

D. COMPARATIVE DEFENSE EVALUATION

Table 16 systematically evaluates nine defense approaches against CIL-DoS, including both established network security mechanisms and our proposed solutions.

Key observations from the comparison:

- **L3/L4 firewalls provide 0% protection** because GOOSE operates at Layer 2. Standard enterprise firewalls cannot even see GOOSE traffic.
- **IDS systems add 100–500** μ**s latency**, consuming 3–16% of the total 3 ms deadline budget. This is unacceptable for inline protection.
- **Switch MAC ACLs are effective but bypassable** through MAC spoofing, which is trivial at Layer 2.

TABLE 16: Comprehensive Comparative Analysis of Defense Approaches Against CIL-DoS Attacks

Defense Approach	Layer	GOOSE-Aware?	Added Latency	Filter Rate	Effectiveness	Primary Limitation
No Defense (Baseline)	–	–	0 μs	N/A	0%	Full CIL-DoS saturation
FIFO + Larger Queue	App	No	0 μs	N/A	0%	Worsens latency proportionally
FIFO + Parallel Workers	App	No	0 μs	$N \times \mu$	0%	Cannot achieve $\rho < 1$
Switch MAC ACL	L2	No	$<$1 μs	Wire speed	Partial	Defeated by MAC spoofing
VLAN Isolation	L2	No	$<$1 μs	Wire speed	Partial	Same-VLAN compromised IED
L3/L4 Firewall	L3	No	50–200 μs	N/A	0%	Cannot inspect L2 EtherType
Snort/Suricata IDS	L7	Yes	100–500 μs	$\sim$300K	Partial	Consumes 16% of 3 ms budget
PQAC (This Work)	**App**	**Yes**	$<$1 μs	$>$100K	**97.6%**	**Overwhelmed at 500K+**
eBPF/XDP + PQAC	**NIC+App**	**Yes**	$<$0.5 μs	**10M+**	$\sim$100%	**Requires Linux 4.8+**

- **PQAC alone handles up to 100K pps** with $<$3% deadline violation. For extreme rates, the combined eBPF/XDP + PQAC architecture provides comprehensive protection.

X. DISCUSSION

A. PRACTICAL SIGNIFICANCE

The experimental results demonstrate that CIL-DoS poses a realistic and severe threat to critical infrastructure. A commodity laptop generating just 10,000 pps of invalid GOOSE frames—achievable with a simple C program and raw socket—can cause **complete protection system failure** on a subscriber using RSA-2048 authentication. In physical terms, this means a transformer experiencing an overcurrent fault would not receive a trip command within the 3 ms safety window, potentially resulting in:

- Transformer overheating and fire
- Busbar damage from sustained fault current
- Cascading protection failures in interconnected substations
- Extended blackouts affecting thousands of customers

B. WHY STANDARD COUNTERMEASURES FAIL

The experimental evidence (Fig. 4) conclusively demonstrates that CIL-DoS exploits a *structural* vulnerability, not a resource limitation:

- **Larger queues** (E06) increase maximum queueing delay proportionally. The fundamental issue is $\rho \gg 1$, which queue size cannot change.
- **More workers** (E07–E08) reduce per-frame delay but cannot achieve $\rho < 1$ because the required thread count ($>$57) exceeds any practical embedded platform capability.
- **Firewalls** are architecturally incapable of inspecting Layer 2 GOOSE traffic.

The only effective approach is **pre-verification traffic classification**: reducing the effective arrival rate at the verification queue before cryptographic resources are consumed.

C. PQAC DESIGN TRADE-OFFS

The PQAC classifier introduces two known limitations:

- **MAC spoofing resilience:** If an attacker spoofs a known publisher MAC, the initial packets are classified as HIGH priority. However, the rate anomaly detector (Stage 2) identifies the flood behavior within 1 second and downgrades the source to LOW. The combined eBPF/XDP + PQAC architecture provides additional defense through NIC-level rate limiting.
- **Classifier warm-up:** As observed in E10 (85% miss at 10K pps), the rate tracker requires $\sim$1 second to establish baselines. This can be mitigated by deploying a "deny by default" initial policy that treats all unknown sources as suspicious until verified.

D. ALGORITHM SELECTION IMPLICATIONS

The algorithm comparison (Table 10) reveals a fundamental design tension:

- **HMAC-SHA256** is inherently CIL-DoS resistant but requires symmetric key management infrastructure.
- **RSA-2048** enables PKI-based key management but is vulnerable to CIL-DoS without PQAC.
- **ECDSA-P256** offers smaller signatures but even higher verification cost on constrained hardware.

We recommend that substation operators deploying RSA or ECDSA authentication *must* implement PQAC or equivalent pre-verification filtering. For greenfield deployments where key management infrastructure can be established, HMAC-SHA256 provides inherent resistance.

E. OPERATIONAL GUIDELINES

Based on our experimental findings, we propose five operational guidelines for substation security engineers and IED vendors:

1) **Implement Pre-Verification Filtering (Mandatory):** Any IEC 62351-6 deployment using RSA or ECDSA *must* implement frame classification before cryptographic verification. FIFO processing of all incoming frames is architecturally indefensible against CIL-DoS. This filtering can be implemented at the NIC level (eBPF/XDP), at the application level (PQAC), or both.
2) **Monitor Verification Queue Depth:** Queue depth should be treated as a primary security metric, analogous to CPU utilization or network bandwidth. Sustained queue growth ($>$50% capacity for $>$1 second) should trigger immediate alert escalation. Our experiments show that queue saturation is the direct precursor to protection failure.
3) **Select Cryptographic Algorithms Based on Hardware:** Table 17 provides algorithm selection guidance.
4) **Ensure Layer 2 Visibility:** Standard L3/L4 firewalls provide zero protection for GOOSE traffic. Security infras-

TABLE 17: Algorithm Selection Guidance by Hardware Tier

Hardware	Recommended	Rationale
Cortex-M4	HMAC-SHA256	RSA infeasible ($T_v > T_d$)
Cortex-R5	HMAC-SHA256	RSA exceeds deadline
Cortex-A9	RSA + PQAC	RSA viable with PQAC
Atom E3800	RSA + PQAC	Sufficient for PQAC
Desktop/Gateway	RSA or ECDSA	Full flexibility

TABLE 18: Consolidated Results Summary

Metric	FIFO	PQAC
At 100K pps RSA-2048:		
Mean latency	2,318 ms	1.49 ms
P99 latency	2,319 ms	10.44 ms
Deadline miss	100%	2.45%
Improvement	—	1,556×
Countermeasure failures:		
4× queue (16K)	9,264 ms	—
4 workers	629 ms	—
Both still 100% miss	Yes	—
Algorithm comparison at 100K:		
RSA-2048 FIFO miss	100%	—
HMAC-SHA256 FIFO miss	1.01%	—
RSA-2048 PQAC miss	—	2.45%

tructure must support Layer 2 EtherType `0x88B8` inspection. Substation network monitoring should include GOOSE-specific dashboards tracking frame rates per source, `stNum`/`sqNum` progression, and signature verification success rates.

5) **Deploy Defense-in-Depth:** No single defense mechanism provides complete protection. We recommend the layered approach: Switch MAC ACLs + VLAN isolation + eBPF/XDP + PQAC, providing redundant filtering from Layer 2 through the application layer.

F. QUANTITATIVE SUMMARY OF RESULTS

Table 18 consolidates the key quantitative findings across all 15 experiments.

G. IMPLICATIONS FOR IEC STANDARDS

Our findings have direct implications for IEC standards development:

- **IEC 62351-6 Amendment:** The current standard specifies authentication mechanisms but does not mandate queue management or pre-verification filtering. We recommend an amendment requiring IED vendors to implement admission control as part of the authentication pipeline.
- **IEC 61850 Performance Requirements:** The 3 ms deadline assumes benign network conditions. Under adversarial conditions, the effective deadline budget consumed by queue waiting reduces the time available for actual verification and protection logic. Standards should specify security-adjusted performance requirements.
- **IED Certification:** IED certification testing should include CIL-DoS resilience benchmarks. A relay that passes functional testing but fails under 10K pps of invalid traffic should not be certified for deployment in critical infrastructure.

H. COMPARISON WITH RELATED APPROACHES

Compared to existing GOOSE security solutions, our approach offers several advantages:

- Unlike IDS-based approaches [10], PQAC operates *before* verification with $<1\ \mu$s overhead, compared to 100–500 μs for deep packet inspection.
- Unlike static MAC filtering, PQAC's rate anomaly detector provides a secondary defense against MAC spoofing attacks.
- Unlike proposals requiring hardware security modules (HSMs) or FPGA-accelerated verification, PQAC is a pure software solution deployable on existing IED hardware without modifications.
- Unlike network-level solutions (firewalls, VLANs), PQAC operates within the IED itself, providing last-line-of-defense protection regardless of network topology.

I. WHY NOT EDF, WFQ, DRR, OR CODEL?

A natural question is whether established network scheduling algorithms could replace PQAC. We argue they are fundamentally unsuitable for the CIL-DoS problem:

- **Earliest Deadline First (EDF):** EDF assumes all tasks are legitimate and schedules by urgency. In CIL-DoS, attack frames carry the same protocol-level urgency as legitimate frames (same `appID`, same `datSet`). EDF cannot distinguish attacker from defender—it would schedule attack frames with equal priority.
- **Weighted Fair Queueing (WFQ):** WFQ allocates bandwidth fairly among flows. Under CIL-DoS with 100K attack pps vs. 1 legitimate pps, WFQ would allocate ∼50% of verification capacity to the attacker. Since $\rho > 1$ even at 50% attack admission, WFQ does not solve the fundamental overload.
- **Deficit Round Robin (DRR):** Similar to WFQ, DRR provides per-flow fairness. With a single attacker flow and a single legitimate flow, DRR gives 50% to each—still admitting 50K attack frames/second, far exceeding the verification capacity of 1,745 fps.
- **CoDel (Controlled Delay):** CoDel drops packets experiencing excessive sojourn time. While CoDel could bound worst-case latency, it drops *indiscriminately*—legitimate frames with long queue waits are dropped alongside attack frames. CoDel has no mechanism to prioritize known-good sources.

The fundamental distinction is that EDF, WFQ, DRR, and CoDel are designed for *flow fairness* in networks where all traffic is presumed legitimate. CIL-DoS is an *admission control* problem: attack traffic should receive *zero* share of verification resources. PQAC achieves this by classifying traffic *before* admission, dropping 99% of flood traffic and prioritizing known-good sources—a capability absent from all four scheduling alternatives.

J. VALIDITY OF HARDWARE SCALING ANALYSIS

Our hardware scaling projections in Table III use clock-frequency ratios to estimate verification latency across ARM processor tiers. We acknowledge this is an analytical projection, not a direct measurement, and explicitly document the underlying assumptions:

1) **Linear scaling model:** RSA modular exponentiation is compute-bound with minimal memory-access variation across ARM Cortex cores. For single-threaded integer workloads, performance scales approximately linearly with clock frequency within the same architecture family [9]. This linear relationship is further corroborated by ARM's own Cortex-A series benchmark data [24], which demonstrates <5% deviation from linear scaling for integer-intensive cryptographic operations across Cortex-A9, A15, and A53 cores.
2) **ISA correction:** Cortex-M4 lacks hardware integer divide and has a simpler pipeline than Cortex-A9, reducing instructions-per-cycle (IPC) by approximately 30–40%. Our scaling factors incorporate this IPC deficit, resulting in Cortex-M4 projections 12× slower than Apple M2 rather than the 21× predicted by pure clock ratio.
3) **Validation reference:** Ali and Choi [9] reported RSA-2048 verification of 845 μs on Cortex-A9 at 800 MHz. Our Cortex-A9 projection of 845 μs (Table III) matches their measurement exactly, validating the scaling methodology for at least one hardware tier.
4) **Conservative bias:** Our projections likely *underestimate* real-world latency because they do not account for OS scheduling jitter, cache contention from concurrent IED tasks, or thermal throttling. Actual hardware measurements would likely show *worse* performance, strengthening our CIL-DoS vulnerability argument.

We emphasize that these are *analytical projections based on published benchmark scaling factors rather than direct measurements on target hardware*, used to assess vulnerability feasibility across IED classes. The specific scaling ratios (8×, 25×, 80×, 300×) derive from clock frequency ratios adjusted for ISA-specific IPC differences. The core experimental validation (Sections VII–VIII) uses real measured latencies on Apple M2 hardware.

XI. LIMITATIONS

We acknowledge the following limitations:

1) **Software testbed:** Experiments were conducted in Docker containers on desktop hardware. Hardware relay timing on ARM Cortex-M4/R5 processors would exhibit *worse* vulnerability due to lower clock speeds, but specific timing characteristics may differ.
2) **Static classifier thresholds:** While our sensitivity analysis (Section VIII-G) demonstrates robustness across $T_{\text{flood}} \in [20, 2000]$, the current PQAC uses fixed thresholds. Adaptive threshold tuning based on real-time traffic profiling could further improve performance across heterogeneous deployment environments.
3) **MAC spoofing at sub-flood rates:** Our MAC spoofing experiments (Section VIII-I) reveal that an attacker using a spoofed MAC at a carefully tuned rate below T_{flood} (e.g., 10K pps) can evade flood classification and degrade performance to 77% deadline miss. This motivates the layered eBPF/XDP defense, which rate-limits per-source before PQAC classification.
4) **MAC allowlist maintenance:** The allowlist is populated dynamically from verified frames. Initial deployment requires manual configuration or a secure bootstrapping protocol. Stale entries must be managed through periodic expiration.
5) **Extreme attack rates:** PQAC alone is insufficient at 500K+ pps, where even 1% admission exceeds the verification rate. The full layered architecture (eBPF/XDP + PQAC) is required for such scenarios.
6) **Single-subscriber evaluation:** We evaluated a single subscriber IED. Multi-subscriber environments with shared process bus may exhibit different traffic patterns and interference effects.
7) **Constant-time verification assumption:** Our RSA implementation uses constant-time verification (∼573 μs) to prevent timing side-channels. Deployments using standard (non-constant-time) RSA (∼34 μs) would have higher saturation thresholds but remain vulnerable at higher attack rates.
8) **eBPF/XDP platform requirement:** The NIC-level defense requires Linux kernel 4.8+ with BPF-capable network drivers, which may not be available on all IED operating systems (e.g., VxWorks, QNX).

A. THREATS TO VALIDITY

We systematically address threats to the validity of our empirical and theoretical findings:

1) **Construct Validity:** Construct validity concerns whether our evaluation metrics accurately reflect operational power system safety. We use the IEC 61850 Type 1A Class P1/P2/P3 hard real-time latency thresholds (3 ms) and queue drop ratios as primary indicators. Because protection relays operate on deterministic trip decisions, exceeding 3 ms directly correlates with delayed circuit breaker tripping and potential transformer or busbar damage during physical overcurrent faults. Furthermore, our testbed implements constant-time cryptographic verification routines to eliminate synthetic timing short-circuits on malformed signatures, ensuring realistic workload stress.
2) **Internal Validity:** Internal validity concerns potential confounding factors within the testbed environment. All experiments were conducted in containerized network namespaces using dedicated virtual Ethernet pairs and isolated CPU affinity to prevent CPU scheduling artifacts. To eliminate cross-experiment state contamination, every run executed in a clean container instance following an automated teardown, volume purge, and 3-second cooldown protocol. Furthermore, all results represent continuous

measurement across thousands of frames with 95% confidence interval reporting.

3) **External Validity:** External validity concerns the generalizability of our findings to operational utility substations. While our experiments executed on an Apple Silicon testbed, our mathematical queueing formulations (Section V) are hardware-agnostic. We mapped operational service rates across five industrial relay tiers using validated processor scaling models. Furthermore, our PQC implementation uses official NIST-standardized C implementations from `liboqs`, ensuring that measured cryptographic characteristics directly reflect production-grade post-quantum algorithms across diverse vendor hardware.

XII. FUTURE WORK

Several research directions emerge from this work:

1) **Adaptive Threshold Tuning:** While our sensitivity analysis (Section VIII-G) demonstrates robustness across a wide range of static thresholds, developing online learning algorithms that automatically adjust PQAC thresholds based on observed traffic patterns would further improve deployment flexibility.
2) **Hardware Relay Validation:** Deploy the GridPulse testbed on actual protection relay hardware (e.g., ARM Cortex-A9 platforms) to validate the vulnerability severity and defense effectiveness under realistic hardware constraints.
3) **Multi-Subscriber Coordination:** Extend the testbed to include multiple subscriber IEDs sharing a process bus, evaluating cross-subscriber interference and coordinated PQAC defense during simultaneous CIL-DoS attacks.
4) **Machine Learning Classification:** Investigate the use of lightweight ML models (e.g., decision trees, online SVMs) for frame classification that can adapt to novel attack patterns beyond rate-based thresholds and MAC allowlists.
5) **IEC Standards Contribution:** Propose amendments to IEC 62351-6 incorporating mandatory pre-verification filtering requirements and queue management guidelines for IED vendors.

XIII. CONCLUSION

This paper presents a comprehensive analysis of Crypto-Induced Latency Denial of Service (CIL-DoS) in IEC 61850 digital substations secured with IEC 62351-6 cryptographic authentication. Through formal mathematical modeling and 32 controlled experiments across multiple configurations, we demonstrate six key findings:

1) **CIL-DoS is devastatingly effective:** Attack rates as low as 10,000 pps cause 100% deadline failure with RSA-2048, producing mean latencies of 2,318 ms (773× the 3 ms safety deadline).
2) **Larger queues worsen the problem:** A 4× increase in queue capacity produces 4× higher latency (9,264 ms) with no improvement in deadline compliance.
3) **Parallel verification is insufficient:** Four verification workers still yield 100% deadline failure because $\rho = 14.3 \gg 1$.
4) **Priority Queue Admission Control is effective:** PQAC reduces deadline violations from 100% to 2.45% and mean latency from 2,318 ms to 1.49 ms at 100K pps—a 1,556× improvement—using the same computational resources.
5) **PQAC is robust and scalable:** Threshold sensitivity analysis across seven T_{flood} values confirms <6% deadline miss at all settings, and multi-publisher experiments demonstrate stable performance with up to 16 concurrent publishers.
6) **Layered defense provides comprehensive protection:** Combining eBPF/XDP NIC-level filtering with PQAC addresses MAC spoofing attacks that bypass application-layer classification alone.

Our results demonstrate that the computational cost of cryptographic authentication in IEC 62351-6 creates a fundamental vulnerability that must be addressed through pre-verification traffic management. We recommend that all IEC 62351-6 deployments using asymmetric cryptography (RSA, ECDSA) implement mandatory pre-verification admission control to maintain real-time protection integrity.

All source code, experiment configurations, timing data, and analysis scripts are released as part of the open-source GridPulse project[1] for full reproducibility.

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